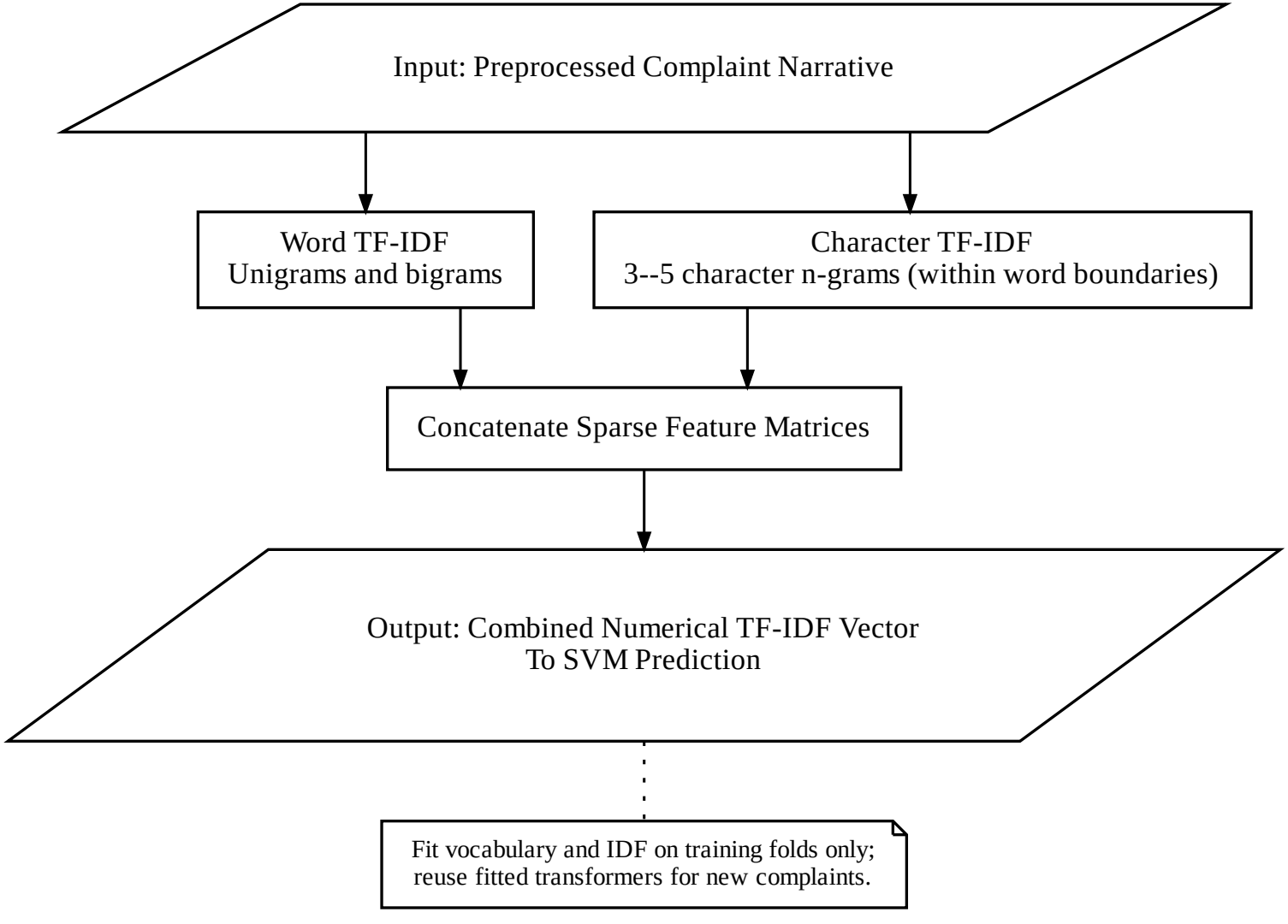
Input: Preprocessed Complaint Narrative
Word TF-IDF
Unigrams and bigrams
Character TF-IDF
3--5 character n-grams (within word boundaries)
Concatenate Sparse Feature Matrices
Output: Combined Numerical TF-IDF Vector
To SVM Prediction
Fit vocabulary and IDF on training folds only;
reuse fitted transformers for new complaints.